

Qualification Pack



Junior Plumber (City Gas Distribution)

QP Code: HYC/Q6106

Version: 1.0

NSQF Level: 2.5

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Qualification Pack

Contents

HYC/Q6106: Junior Plumber (City Gas Distribution)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
HYC/N6114: Carry out helping Activities for Plumbing Work	5
HYC/N6115: Maintaining Plumbing Tools and Equipment	13
HYC/N9301: Working effectively in a team	20
HYC/N9302: Maintain health, safety and security procedures	24
DGT/VSQ/N0101: Employability Skills (30 Hours)	29
Assessment Guidelines and Weightage	34
<i>Assessment Guidelines</i>	34
<i>Assessment Weightage</i>	35
Acronyms	36
Glossary	38

Qualification Pack

HYC/Q6106: Junior Plumber (City Gas Distribution)

Brief Job Description

Plumbing activities in the oil and gas sector involve specialized tasks that differ significantly from residential or commercial plumbing due to the industry's unique requirements and challenges. These activities focus on ensuring the safe and efficient transport and handling of various fluids, including hydrocarbons, water, and chemical solutions, under high pressure and often in hazardous environments. The Helper Plumber assists the head plumber in the installation, maintenance, and repair of plumbing systems. This role involves helping with the assembly, installation, and repair of pipes, fittings, and fixtures of heating, water, and drainage systems, according to specifications and plumbing codes. The helper also organises and hands over tools, clears work area, cuts and bends pipes as per the specified dimensions. The individual also supports the plumber with other tasks as per instructions received.

Personal Attributes

The individual must have a basic knowledge of plumbing systems and materials with an ability to use plumbing tools and equipment. The helper must be able to perform laborious work, be good at receiving and following instructions, have physical stamina and ability to work in various environments (e.g., confined spaces, outdoor settings).

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N6114: Carry out helping Activities for Plumbing Work](#)
2. [HYC/N6115: Maintaining Plumbing Tools and Equipment](#)
3. [HYC/N9301: Working effectively in a team](#)
4. [HYC/N9302: Maintain health, safety and security procedures](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Pipe Fitting in Oil and Gas
Country	India

Qualification Pack

NSQF Level	2.5
Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7126.0301
Minimum Educational Qualification & Experience	9th grade pass with 1-2 Years of experience OR 8th grade pass with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	26/08/2027
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-2.5-HY-02961-2024-V1-HSSC
NQR Version	1.0

Qualification Pack

HYC/N6114: Carry out helping Activities for Plumbing Work

Description

This unit describes how a helper plumber provides support to the main plumber in carrying out installation, repair and maintenance work in an oil and gas sector.

Scope

The scope covers the following :

- Prepare for plumbing work
- Providing help in plumbing work

Elements and Performance Criteria

Prepare for plumbing work

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that all necessary tools and materials are on-site and organized
- PC2.** Set up ladders, scaffolding, and workstations as required.
- PC3.** Transport pipes, fittings, and tools to the work area.
- PC4.** Unload and organize materials from delivery trucks.
- PC5.** Understand and follow local building codes and regulations.
- PC6.** Clear and clean the work area.
- PC7.** Drill holes in walls, floors, and ceilings to accommodate pipes.
- PC8.** Use hangers and brackets to secure pipes in place
- PC9.** Store tools and unused materials neatly.
- PC10.** Read blueprints and drawings to determine exact specifications and where plumbing system will be installed.
- PC11.** Determine water supply networks and waste drainage systems.
- PC12.** Coordinate with project manager on installation.
- PC13.** Install systems used to transport water, waste, gases, or liquids.
- PC14.** Select all materials needed for installation, including type and size of pipe.
- PC15.** unpack pipes, fittings and fixtures as per manufacturer's guidelines and safety measures
- PC16.** check quantity and quality of various plumbing materials and equipment present at the work site
- PC17.** Follow safety protocols and wear appropriate personal protective equipment (PPE) such as gloves, goggles, and helmets.
- PC18.** Clean the work area after installations or repairs are complete.
- PC19.** report any shortcomings of supplies against specifications or requirements

Providing help in plumbing work

To be competent, the user/individual on the job must be able to:

- PC20.** follow the instructions received from head plumber or supervisor

Qualification Pack

- PC21.** Provide help in the installation of pipes and fixtures.
- PC22.** Assist in measuring, cutting, threading, and assembling new pipes.
- PC23.** Support in disassembling and removing damaged pipes or fixtures.
- PC24.** Install supports and hangers for pipe, fixtures and equipment.
- PC25.** Place the assembled pipes in hangers or other supports.
- PC26.** Assemble and install valves and fittings.
- PC27.** Cut openings in structures in preparation for pipes.
- PC28.** Cut pipes to required lengths using pipe cutters, saws, and pipe-cutting machines
- PC29.** Assist in diagnosing problems such as leaks or blockages.
- PC30.** Use basic tools to help locate the source of issues.
- PC31.** Help with pressure tests and other diagnostic procedures to ensure system integrity.
- PC32.** Assist in installing and maintaining plumbing systems.
- PC33.** Apply adhesives, sealants, and caulking to joints and pipe fittings.
- PC34.** Help install plumbing fixtures such as sinks, toilets, and showers.
- PC35.** Position fixtures and ensure they are level and secure
- PC36.** Connect fixtures to water and drainage lines under supervision.
- PC37.** Assist in testing installed systems for leaks and proper operation.
- PC38.** Assist in the maintenance and repair of plumbing systems and fixtures.
- PC39.** Assist in routing pipes through walls, floors, and ceilings
- PC40.** close or release water supply to the fixture and/or pipe being installed or repaired as per instructions of the plumber
- PC41.** use hand and power tools for cutting, bending and assembling pipes and fittings in a safe manner
- PC42.** perform cutting, bending and forming of different types of plastic, GI & SS pipes as per the dimensions marked by the plumber
- PC43.** assemble the fixture and/or pipework as per given instructions
- PC44.** secure the assembly using the fittings and fasteners as per instructions
- PC45.** hold the pipework or fixture steadily during installation for the plumber as and when required
- PC46.** Address basic customer questions and refer more complex inquiries to the supervising plumber.
- PC47.** clear work area and store tools and other plumbing materials safely and as per the specified guidelines
- PC48.** Assist in maintaining records of materials used and tasks completed.
- PC49.** Help with inventory management by tracking materials and tools used during projects.
- PC50.** Communicate with clients to provide updates on work progress.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic knowledge of plumbing systems and materials.
- KU2.** Ability to use plumbing tools and equipment.

Qualification Pack

- KU3.** basic measurement and cutting techniques
- KU4.** organisational quality procedures and processes
- KU5.** organisation's policies on delivery standards, safety and hazards, integrity, dress code, etc.
- KU6.** individual's role in the overall workflow
- KU7.** risk and impact of not following defined procedures/work instructions
- KU8.** international System (SI) of measurement relevant to plumbing work
- KU9.** characteristics and the application of different pipe fittings and fixture supports, including fixing and jointing techniques
- KU10.** process of mains supply of water and drainage
- KU11.** types and grades of pipes, fittings, fastening hardware (such as anchors, studs, bolts, clamps and nuts), supports, restraints, blocking and wall hangers and their properties
- KU12.** installation process of pipes and fittings for various plumbing applications
- KU13.** fixture materials and finishes and uses and limitations of each
- KU14.** corrosion protection ways such as applying coatings and using tape etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions.
- GS2.** Communicate effectively
- GS3.** work in various environments (e.g., confined spaces, outdoor settings)
- GS4.** write common plumbing symbols and brief notes in the local language and preferably Hindi
- GS5.** read and correctly interpret plumbing symbols and notes in local language and English
- GS6.** interpret verbal instructions and information about work related matters correctly
- GS7.** Raise doubts and concerns about work related matters to co-workers and supervisors with clarity
- GS8.** organize tools, materials and equipment on the basis of safety, availability, cleanliness, damage etc.
- GS9.** plan work activities and communicate to others whose work plans and timelines may be affected

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for plumbing work</i>	19	38	-	-
PC1. Ensure that all necessary tools and materials are on-site and organized	1	2	-	-
PC2. Set up ladders, scaffolding, and workstations as required.	1	2	-	-
PC3. Transport pipes, fittings, and tools to the work area.	1	2	-	-
PC4. Unload and organize materials from delivery trucks.	1	2	-	-
PC5. Understand and follow local building codes and regulations.	1	2	-	-
PC6. Clear and clean the work area.	1	2	-	-
PC7. Drill holes in walls, floors, and ceilings to accommodate pipes.	1	2	-	-
PC8. Use hangers and brackets to secure pipes in place	1	2	-	-
PC9. Store tools and unused materials neatly.	1	2	-	-
PC10. Read blueprints and drawings to determine exact specifications and where plumbing system will be installed.	1	2	-	-
PC11. Determine water supply networks and waste drainage systems.	1	2	-	-
PC12. Coordinate with project manager on installation.	1	2	-	-
PC13. Install systems used to transport water, waste, gases, or liquids.	1	2	-	-
PC14. Select all materials needed for installation, including type and size of pipe.	1	2	-	-
PC15. unpack pipes, fittings and fixtures as per manufacturer's guidelines and safety measures	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. check quantity and quality of various plumbing materials and equipment present at the work site	1	2	-	-
PC17. Follow safety protocols and wear appropriate personal protective equipment (PPE) such as gloves, goggles, and helmets.	1	2	-	-
PC18. Clean the work area after installations or repairs are complete.	1	2	-	-
PC19. report any shortcomings of supplies against specifications or requirements	1	2	-	-
<i>Providing help in plumbing work</i>	31	62	-	-
PC20. follow the instructions received from head plumber or supervisor	1	2	-	-
PC21. Provide help in the installation of pipes and fixtures.	1	2	-	-
PC22. Assist in measuring, cutting, threading, and assembling new pipes.	1	2	-	-
PC23. Support in disassembling and removing damaged pipes or fixtures.	1	2	-	-
PC24. Install supports and hangers for pipe, fixtures and equipment.	1	2	-	-
PC25. Place the assembled pipes in hangers or other supports.	1	2	-	-
PC26. Assemble and install valves and fittings.	1	2	-	-
PC27. Cut openings in structures in preparation for pipes.	1	2	-	-
PC28. Cut pipes to required lengths using pipe cutters, saws, and pipe-cutting machines	1	2	-	-
PC29. Assist in diagnosing problems such as leaks or blockages.	1	2	-	-
PC30. Use basic tools to help locate the source of issues.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. Help with pressure tests and other diagnostic procedures to ensure system integrity.	1	2	-	-
PC32. Assist in installing and maintaining plumbing systems.	1	2	-	-
PC33. Apply adhesives, sealants, and caulking to joints and pipe fittings.	1	2	-	-
PC34. Help install plumbing fixtures such as sinks, toilets, and showers.	1	2	-	-
PC35. Position fixtures and ensure they are level and secure	1	2	-	-
PC36. Connect fixtures to water and drainage lines under supervision.	1	2	-	-
PC37. Assist in testing installed systems for leaks and proper operation.	1	2	-	-
PC38. Assist in the maintenance and repair of plumbing systems and fixtures.	1	2	-	-
PC39. Assist in routing pipes through walls, floors, and ceilings	1	2	-	-
PC40. close or release water supply to the fixture and/or pipe being installed or repaired as per instructions of the plumber	1	2	-	-
PC41. use hand and power tools for cutting, bending and assembling pipes and fittings in a safe manner	1	2	-	-
PC42. perform cutting, bending and forming of different types of plastic, GI & SS pipes as per the dimensions marked by the plumber	1	2	-	-
PC43. assemble the fixture and/or pipework as per given instructions	1	2	-	-
PC44. secure the assembly using the fittings and fasteners as per instructions	1	2	-	-
PC45. hold the pipework or fixture steadily during installation for the plumber as and when required	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC46. Address basic customer questions and refer more complex inquiries to the supervising plumber.	1	2	-	-
PC47. clear work area and store tools and other plumbing materials safely and as per the specified guidelines	1	2	-	-
PC48. Assist in maintaining records of materials used and tasks completed.	1	2	-	-
PC49. Help with inventory management by tracking materials and tools used during projects.	1	2	-	-
PC50. Communicate with clients to provide updates on work progress.	1	2	-	-
NOS Total	50	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N6114
NOS Name	Carry out helping Activities for Plumbing Work
Sector	Hydrocarbon
Sub-Sector	
Occupation	Pipe Fitting in Oil and Gas
NSQF Level	2.5
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

HYC/N6115: Maintaining Plumbing Tools and Equipment

Description

This unit deals in detail with the upkeep and maintenance of various plumbing tools and equipment. Regular maintenance of plumbing tools and equipment not only ensures their longevity and reliability but also enhances safety and efficiency on the job.

Scope

The scope covers the following :

- Perform maintenance work

Elements and Performance Criteria

Perform maintenance work

To be competent, the user/individual on the job must be able to:

- PC1.** Clean and maintain plumbing tools and equipment.
- PC2.** Report any damaged or malfunctioning tools to the supervising plumber.
- PC3.** apply suitable treatment method to remove rust and corrosion from damaged tools
- PC4.** Wipe down tools like wrenches, pliers, and pipe cutters with a clean, dry cloth to remove dirt, grime, and moisture after each use.
- PC5.** Clean the exterior of an electrical tools with a damp cloth, avoiding electrical components.
- PC6.** Use compressed air to blow out dust and debris from vents and moving parts.
- PC7.** Inspect tools for any visible damage, such as cracks, chips, or worn-out parts
- PC8.** Perform functionality Test to ensure tools are operating correctly
- PC9.** Check for smooth operation, proper alignment, and effectiveness.
- PC10.** Apply appropriate lubricants to moving parts, such as joints and hinges, to ensure smooth operation and prevent rust.
- PC11.** Use manufacturer-recommended lubricants for power tools, particularly on parts that experience friction.
- PC12.** Perform deep Cleaning for tools that can be taken apart, disassemble them to clean hard-to-reach areas. Use brushes, cloths, and appropriate cleaning solutions.
- PC13.** Apply a rust-preventive solution to metal tools to protect against corrosion.
- PC14.** Sharpen blades, cutters, and other cutting tools using a file, sharpening stone, or an electric sharpener to maintain their effectiveness
- PC15.** Check for signs of wear and tear that might not be immediately visible. Pay attention to handles, grips, and power cords.
- PC16.** Ensure measuring tools and gauges are accurately calibrated.
- PC17.** Replace any missing or damaged tools.
- PC18.** Inspect and clean the motors of power tools, ensuring they are free from dust and debris.
- PC19.** Replace worn-out brushes and other components.

Qualification Pack

- PC20.** Check battery-operated tools, check battery health, clean contacts, and ensure proper charging practices.
- PC21.** Ensure tools are stored in an organized manner in toolboxes, racks, or shelves.
- PC22.** Use protective cases for delicate instruments.
- PC23.** Store tools in a dry, cool place to prevent rust and deterioration.
- PC24.** Ensure tools are used for their intended purposes to avoid unnecessary wear and damage.
- PC25.** Follow manufacturer guidelines and safety instructions.
- PC26.** Wear appropriate PPE (Personal Protective Equipment) such as gloves and safety glasses when cleaning and maintaining tools.
- PC27.** Disconnect power tools from the power source before performing any maintenance.
- PC28.** Provide training to all team members on proper tool usage and maintenance practices.
- PC29.** Keep maintenance logs to track the condition and maintenance schedules of tools.
- PC30.** Perform a comprehensive inventory check to ensure all tools are accounted for and in good condition.
- PC31.** Maintain good inventory of essential supplies like Clean, dry cloths and rags, Brushes and wire brushes, Appropriate cleaning solutions (non-corrosive), Lubricants, Rust-preventive solutions, Sharpening tools (files, sharpening stones), replacement parts (blades, brushes, belts)
- PC32.** Ensure the work area is free of hazards and clean up spills promptly.
- PC33.** Properly dispose of waste materials and debris.
- PC34.** Understand and comply with local building codes and safety regulations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic knowledge of plumbing systems and materials.
- KU2.** Ability to use plumbing tools and equipment.
- KU3.** basic measurement and cutting techniques
- KU4.** organisational quality procedures and processes
- KU5.** organisation's policies on delivery standards, safety and hazards, integrity, dress code, etc.
- KU6.** individual's role in the overall workflow
- KU7.** risk and impact of not following defined procedures/work instructions
- KU8.** international System (SI) of measurement relevant to plumbing work
- KU9.** characteristics and the application of different pipe fittings and fixture supports, including fixing and jointing techniques.
- KU10.** process of mains supply of water and drainage
- KU11.** types and grades of pipes, fittings, fastening hardware (such as anchors, studs, bolts, clamps and nuts), supports, restraints, blocking and wall hangers and their properties
- KU12.** installation process of pipes and fittings for various plumbing applications
- KU13.** fixture materials and finishes and uses and limitations of each
- KU14.** corrosion protection ways such as applying coatings and using tape etc

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions.
- GS2.** Communicate effectively
- GS3.** work in various environments (e.g., confined spaces, outdoor settings)
- GS4.** write common plumbing symbols and brief notes in the local language and preferably Hindi
- GS5.** read and correctly interpret plumbing symbols and notes in local language and English
- GS6.** interpret verbal instructions and information about work related matters correctly
- GS7.** Raise doubts and concerns about work related matters to co-workers and supervisors with clarity
- GS8.** organize tools, materials and equipment on the basis of safety, availability, cleanliness, damage etc.
- GS9.** plan work activities and communicate to others whose work plans and timelines may be affected

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform maintenance work</i>	35	70	-	-
PC1. Clean and maintain plumbing tools and equipment.	1	2	-	-
PC2. Report any damaged or malfunctioning tools to the supervising plumber.	1	2	-	-
PC3. apply suitable treatment method to remove rust and corrosion from damaged tools	1	2	-	-
PC4. Wipe down tools like wrenches, pliers, and pipe cutters with a clean, dry cloth to remove dirt, grime, and moisture after each use.	1	2	-	-
PC5. Clean the exterior of an electrical tools with a damp cloth, avoiding electrical components.	1	2	-	-
PC6. Use compressed air to blow out dust and debris from vents and moving parts.	1	2	-	-
PC7. Inspect tools for any visible damage, such as cracks, chips, or worn-out parts	1	2	-	-
PC8. Perform functionality Test to ensure tools are operating correctly	1	2	-	-
PC9. Check for smooth operation, proper alignment, and effectiveness.	1	2	-	-
PC10. Apply appropriate lubricants to moving parts, such as joints and hinges, to ensure smooth operation and prevent rust.	1	2	-	-
PC11. Use manufacturer-recommended lubricants for power tools, particularly on parts that experience friction.	1	2	-	-
PC12. Perform deep Cleaning for tools that can be taken apart, disassemble them to clean hard-to-reach areas. Use brushes, cloths, and appropriate cleaning solutions.	2	3	-	-
PC13. Apply a rust-preventive solution to metal tools to protect against corrosion.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Sharpen blades, cutters, and other cutting tools using a file, sharpening stone, or an electric sharpener to maintain their effectiveness	1	2	-	-
PC15. Check for signs of wear and tear that might not be immediately visible. Pay attention to handles, grips, and power cords.	1	2	-	-
PC16. Ensure measuring tools and gauges are accurately calibrated.	1	2	-	-
PC17. Replace any missing or damaged tools.	1	3	-	-
PC18. Inspect and clean the motors of power tools, ensuring they are free from dust and debris.	1	2	-	-
PC19. Replace worn-out brushes and other components.	1	2	-	-
PC20. Check battery-operated tools, check battery health, clean contacts, and ensure proper charging practices.	1	2	-	-
PC21. Ensure tools are stored in an organized manner in toolboxes, racks, or shelves.	1	2	-	-
PC22. Use protective cases for delicate instruments.	1	2	-	-
PC23. Store tools in a dry, cool place to prevent rust and deterioration.	1	2	-	-
PC24. Ensure tools are used for their intended purposes to avoid unnecessary wear and damage.	1	2	-	-
PC25. Follow manufacturer guidelines and safety instructions.	1	2	-	-
PC26. Wear appropriate PPE (Personal Protective Equipment) such as gloves and safety glasses when cleaning and maintaining tools.	1	2	-	-
PC27. Disconnect power tools from the power source before performing any maintenance.	1	2	-	-
PC28. Provide training to all team members on proper tool usage and maintenance practices.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. Keep maintenance logs to track the condition and maintenance schedules of tools.	1	2	-	-
PC30. Perform a comprehensive inventory check to ensure all tools are accounted for and in good condition.	1	2	-	-
PC31. Maintain good inventory of essential supplies like Clean, dry cloths and rags, Brushes and wire brushes, Appropriate cleaning solutions (non-corrosive), Lubricants, Rust-preventive solutions, Sharpening tools (files, sharpening stones), replacement parts (blades, brushes, belts)	1	2	-	-
PC32. Ensure the work area is free of hazards and clean up spills promptly.	1	2	-	-
PC33. Properly dispose of waste materials and debris.	1	2	-	-
PC34. Understand and comply with local building codes and safety regulations.	1	2	-	-
NOS Total	35	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N6115
NOS Name	Maintaining Plumbing Tools and Equipment
Sector	Hydrocarbon
Sub-Sector	
Occupation	Pipe Fitting in Oil and Gas
NSQF Level	2.5
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4.5
Credits	1.5
Version	6.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment

Scope

The scope covers the following :

- This unit/task covers the following:
- Practice health and safety measures
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** identify documents, location and people responsible for health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** identify common safety signs, displayed in various areas

Follow fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC7.** use the various appropriate fire extinguishers on different types of fires correctly
- PC8.** follow rescue techniques applied during fire hazard
- PC9.** follow good housekeeping practice in order to prevent fire hazards
- PC10.** list issues concerning the safety in work place
- PC11.** inform fire safety department about any near-miss incidents in the work place
- PC12.** follow the applicable laws, regulations and codes as per safety standard
- PC13.** prepare written accident/incident report and share with the concerned officer/department

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC14.** provide appropriate first aid to victims in emergency situation
- PC15.** demonstrate basic techniques of bandaging
- PC16.** respond promptly and appropriately to an accident

Qualification Pack

- PC17.** Use various rescue equipment & techniques and perform rescue activity during an accident in real or simulated environments
- PC18.** demonstrate correct method to rescue injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks during scaffolding work
- KU5.** health condition of individual as well as health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** working at heights regulations and best practices
- KU9.** various causes of fire
- KU10.** techniques of using different types of fire extinguishers
- KU11.** different materials used for extinguishing fire
- KU12.** various types of safety signs and their significance
- KU13.** various types of rescue equipment and techniques
- KU14.** precautions to be taken to avoid suspension trauma while rescuing a person hanging from harness

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	1	4	-	-
PC2. identify documents, location and people responsible for health and safety in the workplace	2	1	-	-
PC3. identify possible causes of risk or accident in the workplace	1	3	-	-
PC4. carry out safe working practices while dealing with hazards to ensure the safety of self and others	2	2	-	-
PC5. lift heavy objects safely using correct procedures	1	3	-	-
PC6. identify common safety signs, displayed in various areas	2	2	-	-
<i>Follow fire safety procedures</i>	12	16	-	-
PC7. use the various appropriate fire extinguishers on different types of fires correctly	2	2	-	-
PC8. follow rescue techniques applied during fire hazard	2	2	-	-
PC9. follow good housekeeping practice in order to prevent fire hazards	2	2	-	-
PC10. list issues concerning the safety in work place	1	3	-	-
PC11. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC12. follow the applicable laws, regulations and codes as per safety standard	1	3	-	-
PC13. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow emergencies, rescue and first-aid procedures</i>	3	15	-	-
PC14. provide appropriate first aid to victims in emergency situation	1	4	-	-
PC15. demonstrate basic techniques of bandaging	1	4	-	-
PC16. respond promptly and appropriately to an accident	-	3	-	-
PC17. Use various rescue equipment & techniques and perform rescue activity during an accident in real or simulated environments	1	4	-	-
PC18. demonstrate correct method to rescue injured people and others during an emergency	-	-	-	-
NOS Total	24	46	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	3.5
Credits	1.5
Version	5.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 50% of % Aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N6114.Carry out helping Activities for Plumbing Work	50	100	0	0	150	30
HYC/N6115.Maintaining Plumbing Tools and Equipment	35	70	0	0	105	30
HYC/N9301.Working effectively in a team	20	30	-	-	50	10
HYC/N9302.Maintain health, safety and security procedures	24	46	-	-	70	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	149	276	-	-	425	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OS	Occupational Standard(s)
KU	Knowledge and understanding
GS	Generic Skills
SOP	Standard operating procedure
PPE	Personal Protective Equipment
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OS	Occupational Standard(s)
KU	Knowledge and understanding
GS	Generic Skills
SOP	Standard operating procedure
PPE	Personal Protective Equipment
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)

Qualification Pack

NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.